



Methods

Distinctiveness

- In each bigram, phone2's **distinctiveness** w/ phone1 was calculated as the Euclidean distance between their **articulatory/acoustic matrices**.
- The matrices were extracted through self-supervised autoencoders.
- Acoustic/articulatory data → encoder → latent space → decoder → reconstructed data





Methods

Distinctiveness

- In each bigram, phone2's **distinctiveness** w/ phone1 was calculated as the Euclidean distance between their **articulatory/acoustic matrices**.
- Input:
 - Acoustic matrices: the phone token's MFCCs (30 coefficients; 10 frames)
 - Articulatory matrices (EMA/annotated MRI): Temporally normalized coordinates (EMA: 10 frames; annotated MRI: 5 frames)

